

Robin Thibaut, PhD

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SUMMARY

Computational geoscientist specializing in **Bayesian uncertainty quantification** and experimental design for geothermal systems. Build Python workflows that fuse finite-element thermal simulation, geophysics, and well data to **optimize drilling decisions** with **quantified risk**. Publications in *Water Resources Research* and *Journal of Hydrology*; creator of the open-source *SKBEL* framework for Bayesian Evidential Learning.

CORE SKILLS

Programming & ML: Python (NumPy, Pandas, scikit-learn, PyTorch, TensorFlow); Uncertainty Quantification (Bayesian, BEL); Experimental Design; Physics-informed ML; Time-series & Spatial Modeling
Geoscience Modeling: Finite-element thermal simulation; MODFLOW, MT3DMS, MODPATH; ModelMuse; CRTOMO; RES2DINV; SGeMS; Geophysical Data Integration (ERT/IP); Hydrologic & Groundwater Modeling
Data & Engineering: SQL, Snowflake; Git; ETL/Data Engineering; Cloud (Google Cloud Platform); CI/CD; Scientific Visualization; Linux/macOS/Windows

PEER REVIEW & EDITORIAL SERVICE

- Reviewer (2021–2024): *Journal of Hydrology* (4), *Geophysical Journal International* (3), *GEOPHYSICS* (3), *Geophysical Research Letters* (2), *Advances in Water Resources* (4), *Pure and Applied Geophysics* (2), *Hydrogeology Journal* (1), *Water Resources Research* (1), *GEUS Bulletin* (1)

EXPERIENCE

Computational Geoscientist – Zanskar Geothermal & Minerals, USA Oct 2024–Present

- Prioritize drilling targets by fusing finite-element thermal simulations with **ML ranking** to reduce prospect uncertainty and raise decision confidence
- Optimize well placement via **multi-objective search** (thermal drawdown, flow, risk) and **data-driven experimental design**; deliver prioritized drilling recommendations with **techno-economic** screening
- Calibrate 3D finite-element thermal models to well logs and geophysics; **history match** field observations and produce **P10/P50/P90** temperature/flow forecasts for decision support
- Technical lead for the geothermal modeling codebase: own internal repositories, establish code review, unit tests, and CI/CD; manage releases to ensure reproducible modeling across teams
- Operationalize data flows (Python, SQL/Snowflake, Git, GCP) with versioned ETL for site and simulation data, standardizing provenance and enabling dependable reruns
- Ship decision-ready maps and briefs to subsurface teams and leadership, streamlining cross-functional reviews and recommendations

Postdoctoral Researcher – Lawrence Berkeley National Laboratory Aug 2023–Sep 2024

- Integrated geophysical, hydrologic, satellite, and field data into **spatiotemporal ML models** to predict watershed function under climate scenarios
- Built **Python pipelines** (Pandas/NumPy/scikit-learn/PyTorch) for feature engineering and cross-validation; developed reproducible workflows for predictive modeling
- Co-authored model–data integration work on snowmelt/mountain hydrology (*Water Resources Research* 2025); contributed to data collection and analysis

Postdoctoral Researcher – Ghent University (TURBEAMS) Mar 2023–May 2023

- Predicted turbidity/SPM from multibeam backscatter via **deep learning** and point-cloud analytics for water-quality monitoring
- Scaled data handling to millions of points per session, reducing training wall-time

PhD Fellow – Ghent University

Mar 2019–Mar 2023

- Created **Bayesian Evidential Learning** framework for experimental design, selecting measurements that minimize posterior uncertainty and reduce required simulations
- Compared wells vs. geophysics for temperature monitoring within a BEL design framework (WRR 2022), quantifying information-gain vs. cost trade-offs
- Published peer-reviewed articles and presented at international conferences; maintained open-source tools (*SKBEL*, *pysgems*)

Project Engineer – G-tec

Mar 2018–Jan 2019

- Executed marine geophysical surveys (UXO detection, stratigraphy) with QA/QC; delivered coverage on budget and schedule
- Coordinated mobilization and HSE compliance, maintaining an exemplary safety record and improving operational efficiency

SELECTED PUBLIC PROJECTS

Rare Earth Elements — Multiphysics AI-aided Autonomous Prospecting (REE-MAP) Pennsylvania, USA (2023–ongoing)

- Conducted **geophysical data collection, processing, inversion, and interpretation** in a multiphysics, AI-aided approach to identify REE–CM "hot zones" in coal tailings; field-tested at multiple sites; **GSA Connects 2024** abstract with authorship credit and DOI.
GSA Abstract, Paper 404345, DOI: 10.1130/abs/2024AM-404345

TURBEAMS — Towards 3D turbidity by correlating multibeam sonar and in-situ sensor data Belgium (project 2021–2026; contributor 2023)

- Built deep-learning models linking multibeam water-column backscatter with turbidity/SPM; contributed Python data flows toward **3D turbidity/SPM imaging** plus a new **processing library** and workflows.
National research program (New RV Belgica, BELSPO) with published ML objectives (incl. BEL) for large multibeam/optical datasets.

Imaging Saltwater Intrusion, Luy River Coastal Aquifer (Vietnam)

Vietnam (2019–ongoing)

- Co-authored an **open-access** ERT study delineating the saline boundary; **21 ERT profiles** revealed a larger intrusion zone than previously recognized.
Water 2021 (MDPI), DOI: 10.3390/w13131743

Bayesian Evidential Learning (BEL) for Experimental Design in Earth Sciences Belgium (2019–2023)

- Advanced **BEL-based experimental design** to minimize posterior uncertainty and improve monitoring design; resulted in peer-reviewed publications and the **SKBEL** open-source package.
Journal of Hydrology 2021, 10.1016/j.jhydrol.2021.126903; *Water Resources Research* 2022, 10.1029/2022WR033045

EDUCATION

Ph.D. in Geological Sciences – Ghent University

2019–2023

Laboratory for Applied Geology and Hydrogeology

M.Sc. in Geological Engineering – University of Liège

2014–2017

Cum Laude

PUBLICATIONS

- Chen, Hang, **Robin Thibaut**, Chunwei Chou, Chen Xiong, and Yuxin Wu (2025). “A ModEx Framework for Watershed Subsurface Investigation with Limited Geophysical Data Using Machine Learning and Hydrologic Modeling”. In: *Geophysical Research Letters*, *Under review*.
- Tas, Luka, Niels Hartog, Martin Bloemendal, David Simpson, Tanguy Robert, **Robin Thibaut**, Le Zhang, and Thomas Hermans (Jan. 2025). “Efficiency and heat transport processes of low-temperature aquifer thermal energy storage systems: new insights from global sensitivity analyses”. In: *Geothermal Energy* 13. Open access, published 07 January 2025, Article number: 2.
- Wang, Lijing, Zexuan Xu, Chen Wang, **Robin Thibaut**, Craig Ulrich, Matthias Sprenger, Sebastian Uhlemann, Yuxin Wu, Evan King, Haruko Wainwright, Rosemary W. H. Carroll, Curtis Beutler, Kenneth H. Williams, and Baptiste Dafflon (2025). “The Role of Snowmelt and Subsurface Heterogeneity in Headwater Hydrology of a Mountainous Catchment in Colorado: A Model-Data Integration Approach”. In: *Water Resources Research*. DOI: 10.1029/2025WR040651. URL: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR040651>.
- Zhang, Le, Anne-Catherine Dieudonné, Alexandros Daniilidis, Longjun Dong, Wenzhuo Cao, **Robin Thibaut**, Luka Tas, and Thomas Hermans (2025). “Thermo-Hydro-Mechanical Modeling of Geothermal Energy Systems in Deep Mines: Uncertainty Quantification and Design Optimization”. In: *Applied Energy* 377, p. 124531. DOI: 10.1016/j.apenergy.2024.124531. URL: <https://www.sciencedirect.com/science/article/pii/S0306261924019147>.
- Thibaut, Robin**, Ty Ferré, Eric Laloy, and Thomas Hermans (2023). “Sequential optimization of flux measurement to estimate groundwater-surface water interactions”. In: *In preparation*.
- Thibaut, Robin**, Nicolas Compaire, Nolwenn Lesparre, Maximilian Ramgraber, Eric Laloy, and Thomas Hermans (Nov. 2022). “Comparing Well and Geophysical Data for Temperature Monitoring Within a Bayesian Experimental Design Framework”. In: *Water Resources Research* 58 (11). ISSN: 0043-1397. DOI: 10.1029/2022WR033045. URL: <https://onlinelibrary.wiley.com/doi/10.1029/2022WR033045>.
- Cong-Thi, Diep, Linh Pham Dieu, **Robin Thibaut**, Marieke Paepen, Huu Hieu Ho, Frédéric Nguyen, and Thomas Hermans (June 2021). “Imaging the Structure and the Saltwater Intrusion Extent of the Luy River Coastal Aquifer (Binh Thuan, Vietnam) Using Electrical Resistivity Tomography”. In: *Water* 13 (13), p. 1743. ISSN: 2073-4441. DOI: 10.3390/w13131743. URL: <https://www.mdpi.com/2073-4441/13/13/1743>.
- Thibaut, Robin**, Thomas Kremer, Annie Royen, Bun Kim Ngun, Frédéric Nguyen, and Thomas Hermans (Apr. 2021). “A new workflow to incorporate prior information in minimum gradient support (MGS) inversion of electrical resistivity and induced polarization data”. In: *Journal of Applied Geophysics* 187, p. 104286. ISSN: 09269851. DOI: 10.1016/j.jappgeo.2021.104286. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0926985121000331>.
- Thibaut, Robin**, Eric Laloy, and Thomas Hermans (Dec. 2021). “A new framework for experimental design using Bayesian Evidential Learning: The case of wellhead protection area”. In: *Journal of Hydrology* 603, p. 126903. ISSN: 00221694. DOI: 10.1016/j.jhydrol.2021.126903. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0022169421009537>.

SOFTWARE AND DATASETS

- Liang, Zhouji, Sofia Cecilia Brisson, **Robin Thibaut**, Junjie Yu, Carl Hoiland, and Ahinoam Pollack (2024). *Zanskar’s GeoGym*. [Dataset]. URL: <https://www.kaggle.com/datasets/robinthibautzanskar/zanskars-geogym>.
- Deleersnyder, Wouter and **Robin Thibaut** (2022). *Scale-dependent wavelet-based regularization scheme for geophysical 1D inversion*. [Software]. DOI: 10.5281/zenodo.6552695. URL: <http://dx.doi.org/10.5281/zenodo.6552695>.
- Lesparre, Nolwenn, Nicolas Compaire, Thomas Hermans, and **Robin Thibaut** (2022). *4D Temperature Monitoring*. [Dataset]. DOI: 10.34740/kaggle/dsv/3819983. URL: <https://www.kaggle.com/dsv/3819983>.
- Thibaut, Robin** (2021). *WHPA Prediction*. [Dataset]. DOI: 10.34740/kaggle/dsv/2648718. URL: <https://www.kaggle.com/dsv/2648718>.
- Thibaut, Robin** and Maximilian Ramgraber (Sept. 2021). *SKBEL - Bayesian Evidential Learning framework built on top of scikit-learn*. [Software]. Version v2.0.0. DOI: 10.5281/zenodo.6205242. URL: <https://doi.org/10.5281/zenodo.6205242>.

Thibaut, Robin and Guillaume Vandekerckhove (May 2021). *pysgems–Use SGeMS (Stanford Geostatistical Modeling Software) within Python*. [Software]. Version v1.3. DOI: 10.5281/zenodo.4773587. URL: <https://doi.org/10.5281/zenodo.4773587>.

CONFERENCES AND TALKS

- Liang, Zhouji, Sofia Cecilia Brisson, **Robin Thibaut**, Junjie Yu, Carl Hoiland, and Ahinoam Pollack (Feb. 2025). “GeoGym: a Benchmark Dataset for Evaluating Geothermal Exploration Strategies”. In: SGP-TR-229, pp. 1–7. URL: <https://pangea.stanford.edu/ERE/db/GeoConf/papers/SGW/2025/Liang.pdf>.
- Wu, Yuxin, Karthika Balan, Chunwei Chou, Jun Woo Chung, Baptiste Dafflon, Nicola Falco, Jim Panaro, Arun Persaud, Brian Quiter, Emil Rofors, **Robin Thibaut**, Jiannan Wang, Michael L. Whittaker, and John Wu (2024). “RARE EARTH ELEMENTS – MULTIPHYSICS AI-AIDED AUTONOMOUS PROSPECTING (REE -MAP)”. In: 56.5, p. 1. DOI: 10.1130/abs/2024AM-404345.
- Thibaut, Robin**, Ty Ferré, and Thomas Hermans (2023). “Sequential Optimization Of Temperature Measurements To Estimate Groundwater Surface Water Interactions”. In: p. 1.
- Thibaut, Robin**, Nicolas Compaire, Nolwenn Lesparre, Maximilian Ramgraber, Eric Laloy, and Thomas Hermans (2022). “Comparing well and geophysical data for temperature monitoring within a Bayesian Experimental Design framework”. In: p. 20.
- Hermans, Thomas, Nicolas Compaire, **Robin Thibaut**, and Nolwenn Lesparre (2021). “Bayesian evidential learning : an alternative to hydrogeophysical coupled inversion”. In: pp. 3125–3129. ISSN: 1052-3812. URL: <http://dx.doi.org/10.1190/segam2021-3580979.1>.
- Thibaut, Robin**, Thomas Hermans, and Eric Laloy (2021a). “A new framework for experimental design using Bayesian Evidential Learning : the case of wellhead protection area”. In: p. 1.
- (2021b). “Bayesian Evidential Learning combined with experimental design : the case of wellhead protection area prediction”. In: URL: <https://iah2021belgium.org/wp-content/uploads/2021/09/IAH-2021-Book-of-Abstracts.pdf>.
- (2020). “A new framework to reduce uncertainty in Wellhead Protection Area prediction using Bayesian Evidential Learning”. In: p. 1. URL: <https://iemss2020.com/>.
- Thibaut, Robin**, Thomas Kremer, Annie Royen, Bun Kim Ngun, Frederic Nguyen, and Thomas Hermans (2019). “A new approach to incorporate prior information in MGS inversion of ERT/IP data”. In: p. 5. URL: <http://dx.doi.org/10.3997/2214-4609.201902391>.

TEACHING

Teaching Assistant – Groundwater Modeling course, Ghent University	2019–2023
Organizer and Speaker – Python Workshop, Ghent University	Mar 2022
Private Tutor – Mathematics and Physics for university students	2016–2019

LANGUAGES

English (Fluent), French (Fluent), Dutch (Intermediate), Vietnamese (Intermediate)